

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 90%. We keep your highest score.

Next item →

1. Select if the following statements are true or false:

1 / 1 point

True	The softmax function is used to convert logits into probabilities in classification problems.
False	ReLU (Rectified Linear Unit) activation function can produce negative outputs.
True	The sigmoid function maps input values to a range between 0 and 1.
True	Logistic regression is a type of neural network with a single layer.
False	Stochastic Gradient Descent (SGD) updates the weights using the average gradient of the entire dataset.

Response 1

- ☒

 True
- ☐

 False
- ☒

 Nice work

Response 2

- ☐

 True
- ☒

 False
- ☒

 Nice work

Response 3

- ☒

 True
- ☐

 False
- ☒

 Nice work

Response 4

- ☒

 True
- ☐

 False
- ☒

 Nice work

Response 5

- ☐

 True
- ☒

 False
- ☒

 Nice work

2. What is the primary purpose of the activation function in a neural network?

1 / 1 point

- ☐

 To normalize the input data
- ☐

 To initialize weights
- ☐

 To calculate the cost function
- ☒

 To introduce non-linearity into the model

☒

 Nice work

3. Which optimization algorithm is known for updating weights after processing each individual data point?

1 / 1 point

- ☐

 Conjugate Gradient
- ☒

 Stochastic Gradient Descent (SGD)
- ☐

 Gradient Descent
- ☐

 Newton's Method

☒

 Nice work

4. In which type of neural network layer is the ReLU function commonly used?

1 / 1 point

- ☐

 Output Layer
- ☐

 Batch Normalization Layer
- ☐

 Input Layer
- ☒

 Hidden Layer

☒

 Nice work

5. The cost function in logistic regression is also known as:

1 / 1 point

- ☐

 Mean Squared Error
- ☒

 Cross-Entropy Loss
- ☐

 Hinge Loss
- ☐

 Euclidean Distance

☒

 Nice work

6. Which of the following is **NOT** a common type of gradient descent?

1 / 1 point

- ☐

 Batch Gradient Descent
- ☒

 Adaptive Gradient Descent
- ☐

 Mini-batch Gradient Descent
- ☐

 Stochastic Gradient Descent (SGD)

☒

 Nice work

7. The

sigmoid

 function is commonly used in the output layer of a neural network for binary classification.

1 / 1 point

Response 1

- ☐

 ReLU
- ☒

 sigmoid
- ☐

 tanh
- ☐

 softmax

☒

 Nice work

8. In

forward

 propagation, the input is passed through the network to obtain the output.

1 / 1 point

Response 1

- ☒

 forward
- ☐

 reverse
- ☐

 gradient
- ☐

 backward

☒

 Nice work

9. The

regularization

 function helps to prevent overfitting by adding a penalty for large weights in the model.

1 / 1 point

Response 1

- ☐

 activation
- ☐

 cost
- ☒

 regularization
- ☐

 loss

☒

 Nice work

10. The

error

 of a model is typically computed by comparing predicted values with actual values using a cost function.

1 / 1 point

Response 1

- ☐

 weight
- ☒

 error
- ☐

 accuracy
- ☐

 gradient

☒

 Nice work

11. In

mini-batch

 descent, the weights are updated using the average gradient computed from a small batch of training examples.

1 / 1 point

Response 1

- ☒

 mini-batch
- ☐

 stochastic
- ☐

 gradient
- ☐

 batch

☒

 Nice work